Visuospatial representations and looking beyond grammar

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Explorations at the interface between linguistic and visual representations have yielded valuable insights into both language comprehension and production. Visual-world eye-tracking, for example, shows that humans rapidly connect auditory linguistic input to visual context. However, what kind of visuospatial representation does a person construct when they hear/read a sentence in the absence of any relevant visual input? According to the **Spatial Agency Bias** (SAB), humans conceptualize events with Left-to-Right trajectories, with the **left side** associated with agency in languages using left-to-right scripts, e.g. Chatterjee 2001, Maass & Russo 2003, Suitner & Maass 2016, Suitner et al. 2021, Halicki et al. 2023. When asked to draw *The circle pushes the square*, people often draw the agent on the left. (R-L scripts show a right-side privilege, Maass & Russo 2003.) This suggests entities that are more prominent occur in a privileged spatial position.

But researchers disagree on the source and robustness of the SAB. Recently Kaiser 2025 disentangled agentivity relative to other prominence-related concepts, to shed light on the nature of the SAB. By comparing agent-patient verbs (e,g. *chase*) and stimulus-experiencer verbs (e.g. *amaze*) in active and passive voice, Kaiser concludes that both subjecthood and agentivity boost left-side placement and proposes a general *Prominence Bias* instead of an Agency Bias.

However, it is unclear what level this Prominence Bias taps into. To what extent do these asymmetries in visuospatial representations stem from grammatical information vs. information that is not part of core grammatical? All subjects tested by Kaiser were also sentence-initial, so argument structure and linear position were confounded. It is unclear whether effects of subjecthood are simply effects of linear order. Furthermore, it is unclear whether fine-grained semantic distinctions (e.g. intentional human agents vs. forces of nature) and real-world knowledge (e.g. typical/expected agents vs. unexpected agents) also influence the spatial encoding of agentivity. We report three studies testing whether SAB is sensitive to word order, the difference between human agents vs. forces of nature, and typical vs. atypical agents.

Exp1 Clefts. To disentangle linear position and argument structure, **Exp.1** (n=57 native US-English speakers) used **object *it*-clefts**: the lexical verb's *object* was fronted to a focused/clefted position (ex.3). We used the same kind of drawing task as Kaiser (2025): Participants saw sentences one by one, and drew a quick sketch showing the meaning of each sentence (over the internet, using touchpads/mice). Target sentences were transitive; the subject and object were shapes (square, circle, triangle, star, etc) to facilitate subsequent coding of their locations.

The results show an interaction (Fig.1): With stim-exp verbs, whichever argument comes first (subject *or* fronted object) favors left-side placement (above chance, p's<.01). With agent-patient verbs, first-mentioned arguments favor left-side placement *more when they are agents* (SVO) than fronted objects (cleft, p<.05). This shows both agentivity and linear order/focusing matter.

Exp2 Natural forces. To test whether SAB effects are sensitive to finer-grained agentivity distinctions, Exp.2 (n=36) used the same drawing task to compare sentences with human agents vs. forces of nature (e.g. *storm, wave, wind gust, hurricane*), ex.(2). Verbs were agent-patient verbs (e.g. *toppled, rattled)*. The results (Fig.2) show that **both agent types pattern alike**, and replicate Kaiser (2025)'s finding that with agent-patient verbs, subjects favor left-side placement *more when they are agents* (active) than promoted patients competing against demoted *by*-phrase agents (passive, p<.05). Thus, the lack of intentionality of natural forces has no effect, but the behavior of passives shows both subject status and agentivity matter.

Exp3 Unexpected agents. To see if SAB is sensitive to world knowledge, Exp.3 (n=53) tested expected vs. unexpected agents (ex.3). Verbs were agent-patient verbs (e.g. *ambush, interview).* Fig.3 replicates effects of voice and crucially shows agent typicality/expectedness has *no effect*.

As a whole, these studies provide converging evidence for the privileged status of grammatical information (active/passive) over information that also exists outside of grammar (intentionality, real-world knowledge). Our visual representations are highly attuned to grammatical cues.

(1) **Exp 1** (***It*-Cleft**) example (2x2 Latin Square: verb type x SVO/cleft; 24 targets, 24 fillers)

(a) The triangle {tickled$_{\text{AGENT-PATIENT}}$ / amazed$_{\text{STIM-EXP}}$} the circle. [SVO]
(b) It was the triangle that the circle {tickled$_{\text{AGENT-PATIENT}}$ / amazed$_{\text{STIM-EXP}}$}. [it-cleft]

(2) **Exp 2** (**Forces of nature**) example (2x2 Latin Sq: agent type x voice; 24 targets, 24 fil)

(a) {Noah$_{\text{HUMAN}}$ / The tornado$_{\text{FORCE}}$} spilled the bucket of paint. [active voice]
(b) The bucket of paint was spilled by {Noah$_{\text{HUMAN}}$/the tornado$_{\text{FORCE}}$.} [passive voice]

(3) **Exp 3** (**Unexpected agents)** example (2x2 Latin Sq: agent type x voice; 24 targets, 24 fil)

(a) The cheetah stalked the gazelle. [expected agent; active voice]
(b) The gazelle was stalked by the cheetah. [expected agent; passive voice]
(c) The gazelle stalked the cheetah. [unexpected agent; active voice]
(d) The cheetah was stalked by the gazelle. [unexpected agent; passive voice]
(e) Other examples: coyote/deer/bit, leopard/antelope/ambushed, reporter/celebrity/interviewed

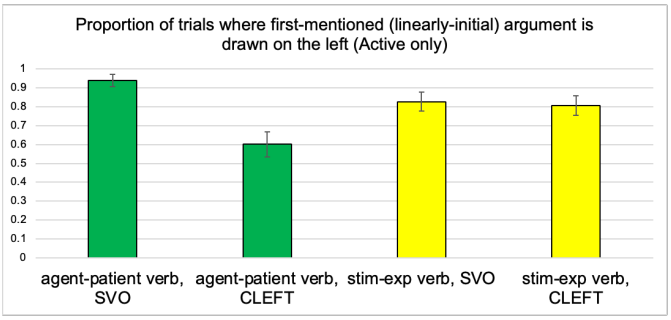

Fig.1 Exp.1: Proportion of trials where the first-mentioned (linearly initial) argument (e.g. *triangle*) is drawn on the left. (Only horizontally-oriented trials included. Error bars: +/- 1 SE)

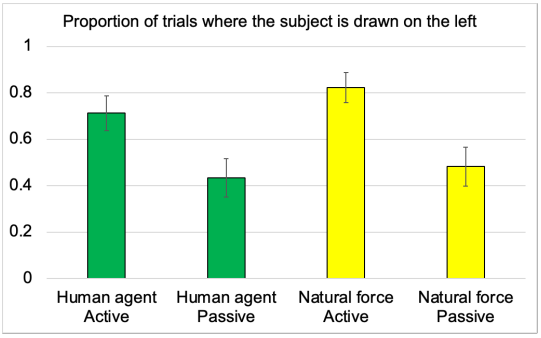

Fig.2 Exp.2: Proportion of trials where the subject is drawn on the left. (Only horizontally-oriented trials included. Error bars: +/- 1 SE)

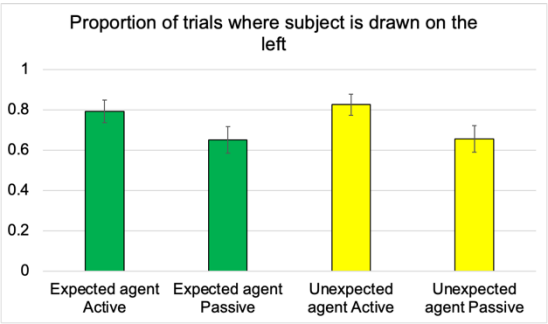

Fig.3 Exp.3: Proportion of trials where the subject is drawn on the left. (Only horizontally-oriented trials included. Error bars: +/- 1 SE)

Examples: The bucket of paint was spilled by the tornado, Monica broke the fence, The gazelle was stalked by the cheetah